Experiment No.3

Objective: Implementation of a 4-bit SIPO and SISO shift registers using flip-flops.

3.1 Equipments Required:

1.	**Non-Consumable**	A. NV6561 Trainer Board B. +5V adaptor C. 2 mm Patch chords
2.	**Consumable**	None

3.2 Basic Principle/Theory involved: A register is a group of binary storage cells suitable for holding binary information. A group of flip-flops constitute a register. Since each flip-flop is a binary cell capable of storing one bit of information, an n-bit register has a group of n flip-flops and is capable of storing any binary information containing n bits. In addition to the flip-flops, a register may have combinational gates that perform certain data processing tasks like when and how new information is transferred in to the register.

One method of describing the shift register characteristics is, by how data is loaded into and is read from the storage units. Depending on this characteristic, shift registers are classified into four types:

- **a.** Serial in-serial out shift register
- **b.** Serial in-parallel out shift register
- **c.** Parallel in-serial out shift register
- **d.** Parallel in-parallel out shift register

3.2.1 Serial in serial out shift register: A basic 4-bit shift register can be constructed using four D flip-flops, as shown in Fig. 3.1. The register is first cleared, forcing all four outputs to zero. The input data is then applied sequentially to the D input of the first flip-flop from the left (FF0). During each clock pulse, one bit is transmitted from left to right. The least significant bit (LSB) of the data is the first to be shifted through the registers i.e. from FF0 to FF3.

In order to get the data out of the register, they must be shifted out serially. This can be done destructively or non-destructively. In a destructive readout, the original data is lost and at the end of the read cycle, all flip-flops are reset to zero. To avoid the loss of data, an arrangement for non-destructive reading can be done by adding two AND gates, an OR gate and an inverter to the system. The construction of this circuit is shown below. The data is loaded to the register when the control line is HIGH (i.e. WRITE). The data can be shifted out of the register when the control line is LOW (i.e. READ).

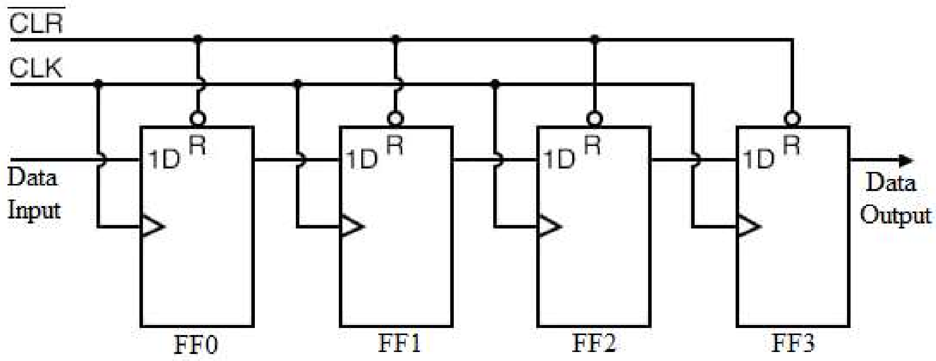

Fig. 3.1 Logic Diagram of Serial in serial out shift register

3.2.2 Serial in parallel out shift register: In this kind of register, data bits are entered serially in the same manner as discussed above. The difference is the way in which the data bits are taken out of the register. Once the data are stored, each bit appears on its respective output line, and all bits are available simultaneously. A construction of a 4-bit serial in parallel out register is shown in Fig. 3.2.

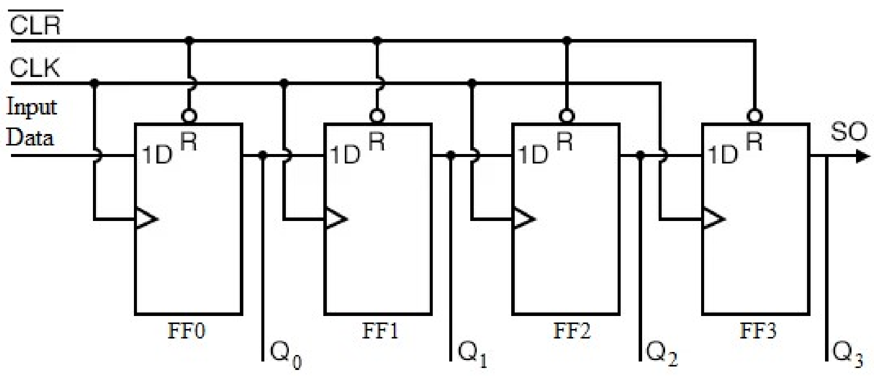

Fig. 3.2 Logic Diagram of Serial in Parallel out shift register

The Q output of a given flip-flop is connected to the D input of the next flip-flop to its right. The serial input determines what goes into the leftmost flip-flop during the shift. Each positive edge of the clock pulse transition shifts the contents of the register one bit position to the right. The serial output is taken from the output of rightmost flip-flop prior to the application of clock pulse. There are four parallel outputs Q0-Q3 with Q3 as LSB. The CLEAR input is an active low input to the flip-flops. It resets or clears the outputs Q0-Q3 when low.

3.3 Procedure:

3.3.1 4-Bit Serial In Serial Out Shift Register: Follow Fig 3.3

Step 1. Connect +5 V adaptor to its indicated position on the trainer board for power supply but do not switch it 'ON'.

Step 2. Connect serial data input D_0 to I_1 of Input section. Toggle it to logic 1 position.

Step 3. Connect the CLK socket using patch cords to socket A of CLK IN. Toggle it to logic 1 level. Clear the outputs Q_0-Q_3 by toggling CLK IN to logic 1 level.

Step 4. Connect PRESET input to I_2 of input section. Toggle it to logic 1 level.

Step 5. Connect CLR input to I0 of input section respectively. Toggle it to logic 0 level

Step 6. Connect the serial data output Q_3 to O_0 of output section.

Step 7. Now switch 'ON' the power supply.

Step 8. Observe output on LED display of O_0 in the output section. It will be off or logic 0.

Step 9. Now toggle CLR input to logic 1 position.

Step 10. Give logic 0 to 1 transition at the CLK IN switch.

Step 11. Observe output on LED display of O_0 in the output section. It will be still off or at logic 0 level.

Step 12. Repeat steps 10-11 and you will observe that the output displays logic 1 or LED glows after 4 positive or 0 to 1 clock transitions.

Step 13. Toggle I_1 connected to D_0 for different inputs from clock pulse number 2-8 as per truth table 1 and verify the truth table.

Step 14. You can also check outputs of all the flip-flops simultaneously for more detailed illustration of the shift operation. You just have to connect the outputs Q_0-Q_3 of flip-flops to the sockets O_0-O_3 in output section.

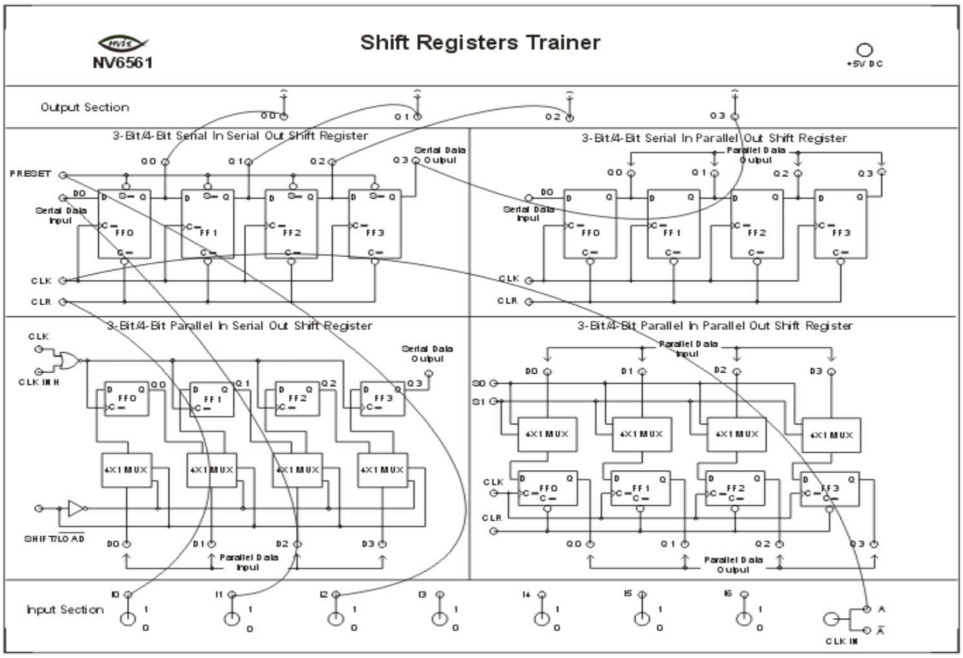

Fig. 3.3 Connection Diagram of 4 Bit Serial in Serial Out Shift Register

3.3.2 4-Bit Serial In Parallel Out Shift Register: Follow Fig 3.4

Step 1. Connect +5 V adaptor to its indicated position on the trainer board for power supply but do not switch it 'ON'.

Step 2. Connect the outputs Q_0-Q_3 of 4-bit serial in parallel out shift register section to O_0- O_3 of output section respectively.

Step 3. Connect serial data input D_0 to I_0 of Input section. Toggle it to logic 1 position.

Step 4. Connect the CLK socket using patch cords to socket A of CLK IN. Toggle it to logic 1 level. Clear the outputs Q_0-Q_3 by toggling CLK IN to logic 1 level.

Step 5. Connect the CLR input to I_1 of input section. Toggle it to logic 0 level.

Step 6. Switch 'ON' the power supply.

Step 7. Observe outputs on LED display of the output section. It will be 0000.

Step 8. Now toggle CLR input through I_1 switch to logic 1 level.

Step 9. Give logic 0 to 1 transition to CLK input.

Step 10. Observe outputs on respective LED displays O_0-O_3 of output section. It will be 1000.

Step 11. Repeat steps 9 and 10 for different inputs from clock pulse number 2-7 as per truth table 3 and verify the truth table.

3.4 Observation Table

Table 3.1: Truth table for 4-bit serial in serial out shift register

CLOCK PULSE NO.	PRESET	CLR	CLK	D_0	Q_3
0	1	0	1	1	0
1	1	1	0⟶1	1	0
2	1	1	0⟶1	0	0
3	1	1	0⟶1	1	0
4	1	1	0⟶1	0	1
5	1	1	0⟶1	0	1
6	1	1	0⟶1	0	0
7	1	1	0⟶1	0	1
8	1	1	0⟶1	0	0

Table 3.2: Truth table for 4-bit serial In parallel out shift register

CLOCK PULSE NO.	CLR	CLK	D_0	Q_0	Q_1	Q_2	Q_3
0	0	1	1 or 0	0	0	0	0
1	1	0⟶1	1	1	0	0	0
2	1	0⟶1	0	0	1	0	0
3	1	0⟶1	1	1	0	1	0
4	1	0⟶1	0	0	1	0	1
5	1	0⟶1	0	0	0	1	0
6	1	0⟶1	0	0	0	0	1
7	1	0⟶1	0	0	0	0	0

3.5 Result: 4-bit SIPO and SISO shift registers has been implemented and operation is verified.

3.6 Precautions:

I. Connection should be done properly.
II. Don't make connections with wet hands.
III. While doing connections main power supply should be off.

3.7 Viva Voce questions:

1. What does serial in serial out mean?

2. How does serial shift register work?

3. How does serial in parallel out shift register work?

4. How many clock pulses are required to completely load serially a 8-bit shift register?

5. How many clock pulses are required to completely load serially and out parallel, a 8-bit shift register?

6. Which of the following mode is used as control inputs to a 4-bit universal shift register?

a. Loopback mode
b. Parallel load mode
c. Serial input mode
d. All of the above

Answer: Option b

7. The number of flip-flops in a shift register is dependent upon?

a. The modulus of the counter
b. The number of stages in the counter
c. The number of digits to be stored
d. None of the above

Answer: Option c

8. In which mode can we provide data to all the flip-flops simultaneously?

a. Loopback mode
b. Serial input mode
c. Parallel input mode
d. All of the above

Answer: Option b

9. A shift register is a digital circuit that ______________ .

a. Stores data.
b. Shifts the data from left to right.
c. Shifts the data from right to left.
d. all of the above.

Answer: Option d

10. What is a shift register?

a. An adder circuit
b. A memory circuit
c. A combinational circuit
d. A decoder circuit

Answer: Option b

11. What type of register is a shift register?

a. Digital
b. Analog
c. Both 1 and 2
d. Neither 1 nor 2

Answer: Option a

12. A shift register is made up of how many flip-flops?

a. One
b. Two
c. Three or more
d. None of the above

Answer: Option c

13. What happens when a shift register is clocked?

a. Stays the same (no change)
b. Changes to the next bit in sequence (shifts)
c. Changes to an opposite state (complements)
d. None of the above

Answer: Option b

14. A shift register can have __________ data input and __________ data output lines.

a. Only one, only one
b. Two, two
c. One, two
d. Three, two

Answer: Option c

15. A shift register can be used as a ___________-bit register by connecting all the parallel outputs together and applying the same input to all the parallel inputs.

a. 4-bit
b. 8-bit
c. 16-bit
d. n-bit

Answer: Option d

16. Which among the following is the main advantage of using a shift register?

   a. Ease of use and cost effective solution for applications that require several I/O pins at a time.
   b. It is used where single input and single output is used.
   c. Ease of use and cost effective solution for applications that require less number of I/O pins at a time.
   d. None of these.

   Answer: Option a

17. Which of the following type of shift register is used in 8085 microprocessor?

   a. Serial
   b. Parallel
   c. Both a and b
   d. None of the mentioned

   Answer: Option c

18. Which of the following is true about shift registers?

   a. It is not used to store multi-bit data.
   b. They are available only in the parallel mode of operation.
   c. They are useful for data transfer from one location to another.
   d. All of the above.

   Answer: Option c

19. Which of the following is not a characteristic of universal shift registers?

   a. Serial in/serial out (SISO) operation is possible.
   b. SISO operation is not possible.
   c. Serial in/parallel out (SIPO) operation is possible.
   d. Parallel in/serial out (PISO) operation is possible.

   Answer: Option b

20. Which of the following shift registers can be used as both PISO and SIPO?

   a. 4-bit parallel-in serial-out shift register.
   b. 4-bit universal shift register.
   c. 4-bit serial-in/serial-out shift register.
   d. 4-bit parallel load serial in, serial out shift.

   Answer: Option b

21. What is the difference between a SISO, SIPO and PISO shift register?

   a. A SISO has one data input, while a SIPO has 2 .
   b. A SIPO has one data input, while a PISO has 2.
   c. A PISO has one data input, while a SISO has 2.
   d. A PISO has two data inputs, while a SIPO has 1.
   e. None of the above answers are correct.

   Answer: Option e

22. What is one of the main functions of a shift register?

   a. To convert digital information into analog signals.
   b. To control voltage levels according to clock pulses.
   c. To store bits and bytes of binary data temporarily.
   d. To convert serial digital information into parallel or parallel to serial.

   Answer: Option d

23. Which of the following shift registers are also called universal shift registers?

   a. n-bit right shifter with parallel load.
   b. n-bit right shifter with parallel load & clear.
   c. n-bit right shifter and left shift register with parallel load.
   d. All of above.

   Answer: Option c

24. A ________ is a group of special circuit with their outputs connected in series to form a register.

   a. Data register
   b. Shift register
   c. Control register
   d. Accumulator

   Answer: Option b

25. What are the two basic types of shift registers?

   a. Latched & unlatched registers.
   b. 4-bit & 8-bit registers.
   c. SISO & PIPO
   d. SINO & PINO

   Answer: Option c